

Troubleshooting Guide — Symptom Reference

Quick Reference · Organized by Machine Type

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Owner	Maintenance Department
Scope	CNC Milling, Press Line, Assembly Line — all plants

1. Purpose

This guide is organized by observed symptom rather than by downtime code, so operators and technicians can look up a problem the way it is actually reported on the floor. Each row cross-references the relevant downtime code (where one applies) and the SOP or reference document that contains the full corrective procedure. Symptoms with no direct downtime code are quality or condition-monitoring issues that should still be logged in the CMMS even though they may not stop production immediately.

2. CNC Milling

Symptom	Likely Cause	Recommended Action	Code
Chatter marks / rough surface finish	Incorrect feed or speed, dull tool, unstable workholding, or spindle wear	Correct feed/speed to spec, replace worn tooling, reduce tool overhang and improve fixturing, check spindle runout	—
Spindle won't start, stops unexpectedly, or runs hot/noisy	VFD or motor fault, loose drive belt, overheating, blown fuse or tripped breaker	Check power supply and breakers, inspect belts and cooling fan, test drive per Electrical_Troubleshooting_SOP	DT-04
Parts drifting out of dimensional tolerance	Tool offset drift or worn ball-screw backlash	Re-verify tool offsets per Calibration_SOP; inspect ball-screw and linear guide backlash	DT-06
Axis lag alarm or poor servo response	Dry way surfaces, worn couplings, or low servo gain	Grease way surfaces per Lubrication_Schedule, inspect couplings, re-run controller auto-tune	—
Tool changer fails to index or tool won't seat	Worn ATC mechanism or incorrect tool-call sequence in program	Inspect automatic tool changer per Changeover_SOP; verify tool-call sequence in the part program	DT-03
Workholding won't clamp or clamp pressure low	Hydraulic workholding circuit losing pressure	Check clamp pressure gauge; inspect hydraulic clamp circuit per Hydraulic_Maintenance_SOP	DT-01
High spindle vibration or grinding noise	Spindle or main bearing wear	Stop machine, inspect bearing condition, replace per Bearing_Replacement_SOP	DT-08
Repeated short unplanned stops, no visible fault	Sensor or limit switch fault flagged in PLC log	Check PLC fault log for the flagged I/O address; inspect, clean, or replace sensor per Electrical_Troubleshooting_SOP	DT-04

3. Press Line

Symptom	Likely Cause	Recommended Action	Code
Press won't build full tonnage / weak pressing force	Worn cylinder seals, damaged pump plunger, or contaminated check valve	Depressurize and inspect seals, pump, and check valve per Hydraulic_Maintenance_SOP; replace worn parts	DT-01
Oil leaking around piston seal or fittings	Overfilled reservoir, worn packing, or loose fittings	Correct oil level, replace packing/seals, re-torque fittings per Hydraulic_Maintenance_SOP	DT-01
No hydraulic pressure at all	Directional control valve reversed or solenoid valve stuck/dirty	Verify directional control valve setting; clean solenoid valve core per Hydraulic_Maintenance_SOP	DT-01
Hydraulic system running hot	Continuous high-duty operation or restricted cooling flow	Clean cooler/radiator, verify coolant flow, check duty cycle against rated spec	—
Ram/slider descends unevenly or jerky motion	Air trapped in the hydraulic circuit or faulty control valve	Bleed the hydraulic circuit per Hydraulic_Maintenance_SOP; inspect the control valve	DT-01
Excessive changeover time between die setups	Setup steps done with the machine stopped instead of pre-staged externally	Review sequence against Changeover_SOP; shift eligible steps to external (pre-staged) setup	DT-03
Tonnage or pressure gauge reading drifting	Gauge or pressure sensor calibration drift	Verify gauge against certified reference standard per Calibration_SOP	DT-06
Line waiting on incoming blanks/material	Upstream feed or raw material shortage	Escalate to logistics per PM_Guide; check buffer stock level against schedule	DT-02

4. Assembly Line

Symptom	Likely Cause	Recommended Action	Code
Conveyor belt drifting or mistracking to one side	Frame not level/square, worn or misaligned rollers/pulleys, or uneven loading	Check frame level and diagonal squareness, align pulleys and idlers per Inspection_Checklist	DT-05
Frequent jams at transfer points	Material buildup, sharp edges snagging parts, or irregular part orientation	Clear buildup, inspect transfer-point geometry, adjust part feed orientation	DT-05
Rollers seized or hard to turn	Dirt/debris buildup or inadequate lubrication	Clean and re-lubricate rollers per Lubrication_Schedule	DT-05
Unexpected E-stop trip with no obvious cause	Misaligned or faulty safety sensor/light curtain	Inspect, realign, and clean sensor; check wiring per Electrical_Troubleshooting_SOP	DT-04
Torque tool fails calibration check or gives inconsistent readings	Clutch wear, missed calibration interval, or wrong setting selected	Remove from service, recalibrate per Calibration_SOP, confirm correct torque spec is selected	DT-06
Conveyor motor drawing excessive current / overload trip	Worn gear reducer, misaligned coupling, or a jam further down the line	Check for a downstream jam first, then inspect gear reducer and coupling alignment	DT-05
Lift/tilt fixture not actuating or slow to actuate	Shared hydraulic power unit losing pressure	Inspect the shared hydraulic power unit per Hydraulic_Maintenance_SOP	DT-01

Symptom	Likely Cause	Recommended Action	Code
Recurring downtime right at shift start	Incomplete shift handover or briefing	Reinforce shift handover checklist per Plant_Maintenance_Policy	DT-07

5. Escalation Reminder

If the recommended action does not resolve the symptom on the first attempt, or the underlying downtime code is classified Major (DT-01, DT-02, DT-08) in the Error_Code_Handbook, escalate immediately per the Plant_Maintenance_Policy escalation matrix rather than repeating troubleshooting steps.

■ This guide is a triage aid, not a substitute for the full SOP. Always follow lockout-tagout in Safety_Guidelines before any physical inspection or repair.